

 COMPLETE ARCHITECTURE & KNOWLEDGE GUIDE

WalkMate Architecture & Core Concepts

A comprehensive deep-dive into how WalkMate was conceived, architected, and built—from real-time GPS trigonometry and MET exercise science to Japanese editorial Polaroid memory cards.

Platform: Flutter
(Android)

Language:
Dart

Architecture: Service-Oriented Reactive
MVC

Privacy: 100% Zero-Route
Storage

1. Philosophy: Why WalkMate is Different

Modern fitness applications (such as Strava, Nike Run Club, or Apple Fitness) are built around **competition, public leaderboards, pace pressure, and route bragging**. For everyday mindful walkers, this creates anxiety rather than wellness.

WalkMate was built with a fundamentally opposite paradigm:

- **No Destination or Route Required:** You don't need to "plan a track". Step outside your door and take 300 meters or 4 kilometers. Distance is purely cumulative.
- **Absolute Privacy by Architecture:** Most fitness apps secretly upload your breadcrumb GPS route to cloud servers. WalkMate *never* stores coordinates in memory or on disk. Once a delta is computed, the latitude and longitude are immediately dropped.
- **Calming Botanical Aesthetics:** Instead of high-voltage neon red or aggressive athletic blacks, WalkMate uses Japanese forest bathing (*Shinrin-yoku*) tones: Sage Green, Oat Sand, and Peach Blossom.
- **Non-Competitive Milestone Cheers:** Every 500 meters, you receive a gentle word of gratitude and body-mind scientific insights instead of a speed comparison.

The Zero-Route Privacy Guarantee

In WalkMate, the GPS stream only exists during the active walk. The app computes: `delta = distance(previousLat, previousLng, currentLat, currentLng)`. Once `cumulativeDistance += delta` executes, `previousLat/Lng` are overwritten. No list of points, GeoJSON, or map polylines are ever constructed or saved.

2. Flutter Foundations: How the Code Works

To understand WalkMate, you only need to grasp three core concepts of the Flutter framework:

1. "Everything is a Widget"

In Flutter, buttons, text, paddings, animations, and entire screens are all **Widgets** nested together in a tree hierarchy. When data changes, Flutter redraws only the specific sub-tree.

2. Stateless vs. Stateful

`StatelessWidget` is static (like a badge or button).

`StatefulWidget` holds dynamic state (like a timer ticking, weight input, or active walk metrics) via `setState()`.

State Management with `ChangeNotifier`

Instead of using heavy, bloated state management libraries, WalkMate uses Flutter's built-in `ChangeNotifier`. Our `TrackingService` extends `ChangeNotifier`. When GPS moves or the 1-second timer ticks, it calls `notifyListeners()`. The UI screen listens and updates seamlessly at 60 FPS without freezing the main thread.

3. Deep Dive into Every Feature & Concept

3.1 One-Time Weight Persistence (StorageService)

Concept: Why ask for weight? Energy expenditure in walking is directly proportional to body mass. Moving a 90 kg mass 1 kilometer requires significantly more energy than moving a 55 kg mass.

- **Package Used:** `shared_preferences` (Local key-value store using Android's native `SharedPreferences`).
- **Normalized Storage:** Weight is always stored internally in **kilograms (kg)**. If a user enters `154 lbs`, the app converts and stores `70.0 kg`:
`weightKg = displayWeight / 2.20462`.
- **Privacy Reassurance:** Data remains in the application's private sandboxed data directory on the Android device. Zero cloud sync.

3.2 GPS Tracking & Cumulative Distance Math (TrackingService)

Concept: How does temporary GPS tracking work without logging routes?

- **Package:** `geolocator` (interfaces with Android's `FusedLocationProviderClient`).
- **The Distance Filter (Anti-Jitter):** When standing still, GPS coordinates drift naturally by 1–5 meters due to atmospheric interference. If recorded raw, the user would accumulate "ghost distance". WalkMate sets `distanceFilter: 3` meters and requires accuracy under `35m` to accept a coordinate.
- **Spherical Trigonometry (Haversine Formula):** The distance between two GPS points on Earth's curved surface is calculated using the spherical law of cosines:

$$a = \sin^2(\Delta\text{lat}/2) + \cos(\text{lat1}) \cdot \cos(\text{lat2}) \cdot \sin^2(\Delta\text{lon}/2)$$

$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$\text{Distance (meters)} = \text{EarthRadius (6,371,000m)} \cdot c$$

- **Speed & Teleport Filter:** If an incoming GPS point implies speed greater than `18 km/h` (e.g. user got into a car or GPS jumped across cell towers), the delta is ignored to preserve pure walking metrics.

3.3 MET-Based Calorie Estimation (CalorieService)

Concept: Many apps guess calories using generic step counts. WalkMate uses the **Compendium of Physical Activities**, the gold standard in exercise physiology developed by Stanford and the American College of Sports Medicine.

What is a MET? MET stands for *Metabolic Equivalent of Task*. `1.0 MET` is the rate of energy spent sitting quietly at rest (approx. $\$1\text{ kcal} / \text{kg} / \text{hr}\$).$

$$\text{Calories per minute} = (\text{MET} \times 3.5 \times \text{Weight_in_kg}) / 200$$

$$\text{Calories per second} = \text{Calories_per_minute} / 60$$

Walking Intensity	Speed Range (km/h)	MET Value	Physical Description
Resting / Paused	0.0 km/h	1.2	Resting metabolic state during water break
Casual Stroll	< 2.5 km/h	2.0	Window shopping, gentle wandering in garden
Relaxed Walk	2.5 – 3.5 km/h	2.8	Effortless, conversational walking pace
Moderate Walk	3.5 – 4.5 km/h	3.3	Purposeful stroll, standard walking cadence
Brisk Walk	4.5 – 5.5 km/h	3.8	Aerobic pace, active arm swing
Very Brisk / Power	5.5 – 6.5 km/h	4.3	Power walk, deep rhythmic breathing
Fast / Uphill	> 6.5 km/h	5.0	Hurried stride, maximum walking burn

3.4 500m Milestone Cheers & Insights (MilestoneService)

Concept: Traditional trackers wait until 1 km or 5 km to notify you. For a beginner or stressed individual, 1 km can feel far. Every **500 meters (0.5 km)**, WalkMate triggers a celebratory micro-banner:

Milestone	Supportive Message	Body & Mind Insight
0.5 km 	"Nice start! Taking the first step is always the hardest."	Breathing and pulse settling into natural cadence.
1.0 km 	"One full kilometer! Feel the earth gently supporting your stride."	Fresh oxygen circulating to the brain's hippocampus.
1.5 km 	"You've settled into a lovely rhythm. Notice the quiet around you."	Neck and shoulder tightness naturally decompressing.
2.0 km 	"Two kilometers of self-care. Notice how your thoughts clear."	~100 kcal burned: powers 2 hours of sharp cognitive focus.
3.0 km 	"Three full kilometers. Beautiful dedication to your well-being."	Endorphin release creates gentle, lasting mood elevation.
5.0 km 	"Five whole kilometers of peaceful walking! Celebrate this moment."	Over 6,500 mindful steps strengthening cardiovascular health.

3.5 The Breathing Pulse Circle (PulseCircle)

Concept: During an active walk, the center circle doesn't just show distance—it gently expands and contracts in a **3.2-second smooth sine curve**. This timing matches the physiological *box-breathing cadence*, subconsciously guiding the user to breathe deeply and release stress while walking.

3.6 The Polaroid Walk Memory Card (MemoryCardService)

Concept: When you finish a walk, you shouldn't just see a boring spreadsheet of numbers. You created a memory!

- **Editorial Polaroid Layout:** Off-white textured background, subtle warm shadows, rounded photo frame, typography with serif accents.
- **Photo Options:** Take a photo with Camera or pick from Gallery via `image_picker`. If no photo is chosen, it automatically renders a calming botanical tree illustration with sun rays.
- **RepaintBoundary Technology:** Flutter can render any widget subtree directly into an off-screen bitmap at `3.0x pixel ratio` (Retina print quality).
- **Share & Download:** The bitmap is written as a PNG to temporary storage and opened directly in Android's native share tray (WhatsApp, Instagram Stories, Telegram, or Gallery).

4. Project Code Blueprint: What Each File Does

Here is the exact map of the entire WalkMate codebase so you know exactly where everything lives:

File Path	Layer	Core Responsibility
lib/main.dart	App Entry	Initializes Flutter bindings, locks portrait orientation, boots StorageService and launches MaterialApp.
lib/theme/app_colors.dart	Theme	Defines the curated palette: Forest Sage (#3E664F), Mint Mist (#E9F1EC), Oat Sand (#F4F1EA), and Peach (#E58F65).
lib/theme/app_theme.dart	Theme	Configures Material 3 ThemeData with Google Fonts (Outfit), soft card elevations, and rounded button borders.
lib/models/walk_session.dart	Model	Stores summary stats (distance, duration, calories, avg speed, photo, phrase). Never stores GPS points.
lib/models/milestone_info.dart	Model	Holds milestone distance, title, supportive cheer, insight text, and emoji icon.
lib/models/movement_insight.dart	Model	Collection of scientific calorie, movement, and resting insights.
lib/services/storage_service.dart	Service	Interfaces with SharedPreferences for saving user weight (kg/lbs) and lifetime walk summaries.
lib/services/calorie_service.dart	Service	Calculates MET values based on walking speed tiers and computes real-time calories burned.
lib/services/tracking_service.dart	Service	Manages GPS location stream, distance accumulation, speed smoothing, pause/resume, and milestone detection.
lib/services/milestone_service.dart	Service	Detects when cumulative distance crosses 500m boundaries and returns the appropriate encouraging message.
lib/services/memory_card_service.dart	Service	Captures RepaintBoundary widget tree as high-res PNG image and triggers native Android share sheet.
lib/widgets/weight_dialog.dart	Widget	Modal dialog for entering weight once with interactive kg/lbs toggle and privacy reassurance.
lib/widgets/pulse_circle.dart	Widget	Concentric animated breathing circle creating a soothing pacing effect on the active tracking screen.
lib/widgets/milestone_banner.dart	Widget	Celebratory banner card that pops in whenever a 500m milestone is achieved.
lib/widgets/stat_badge.dart	Widget	Reusable metric tile with icon, title, large numeric value, and unit.
lib/widgets/walk_memory_card.dart	Widget	The visual centerpiece: Polaroid editorial memory card with photo frame, stats, quote, and watermark.
lib/screens/home_screen.dart	Screen	Welcoming dashboard with time-of-day greeting, weight badge, Start Walk button, and lifetime stats.

File Path	Layer	Core Responsibility
lib/screens/active_walk_screen.dart	Screen	Real-time walk tracking screen with pulse circle, live metrics, milestone toasts, and pause/finish controls.
lib/screens/walk_summary_screen.dart	Screen	Summary screen showing the live Walk Memory Card preview, camera photo picker, phrase picker, and share button.
android/.../AndroidManifest.xml	Android Native	Declares location permissions (FINE/COARSE), camera access, and foreground service permissions.

5. How to Run, Test, and Deploy

A. Live Interactive Simulator (Already Running!)

You can run and test all screens, walk simulations, 500m milestones, and memory card exports right now in your browser at:

`http://localhost:3456`

B. Running in Android Studio

1. Open **Android Studio**.
2. Click **Open** and choose the folder: `Runwithme`.
3. In the device selector at the top bar, pick your installed emulator (**Pixel_4** or **Medium_Phone_API_36**).
4. Click the green **Run ▶** button. Android Studio will automatically build the debug APK and launch it on your virtual phone!

C. Testing on Your Physical Android Phone

1. On your Android phone, go to **Settings > About Phone** and tap **Build Number** 7 times to enable Developer Options.
2. Go to **Developer Options** and toggle **USB Debugging** to ON.
3. Plug your phone into your PC via USB cable and allow the debugging prompt on your phone screen.
4. In terminal, run: `flutter run`. The app installs directly onto your phone!

You Built WalkMate!

Every single line of this application was designed specifically for your vision: a supportive, private, non-competitive companion that honors your movement. You now understand every formula, widget, and architectural decision inside out.